



PEAS Scheme of Work: Biology

CLASS: S.4

peas
PROMOTING EQUALITY IN AFRICAN SCHOOLS

SUBJECT: BIOLOGY **CLASS** S.4 Term: 1 NAME OF
TEACHER:

Time allocation: 4 periods per week (2 doubles)

When it comes to planning your lessons use the structure: starter, I do, we do, you do, plenary. The starter is a written task that reviews previous learning. Ensure your lessons provide regular and extended opportunities for independent practice.

YPR:

Y= yes, I taught the lesson

P= I partially taught it e.g. I didn't get through all the content.

R= I taught the lesson, but I think students would benefit from a review

Week&Lesson Subtopics	Learning outcome(s)	Methodology	Teaching/ Learning resources	Y P R
Theme: Reproduction in organisms Topic: Asexual reproduction in plants. Competency: <i>The learner appreciates that some parts of plants can develop in new independent plants</i>				
1.1 Asexual reproduction	<i>-knows the meaning of asexual reproduction</i>	- Asks Learners to move out of class carryout study in school compound or around it to observe plants, identifying possible parts that have been or could have been used to get new plants of the same kind. - Informs learners share their ideas to classmates and teacher. - Individual Learners report their findings in a table	Fountain biology book 4 page 1 St. Bernard biology book 4 page 1.	

		with drawings as appropriate.		
1.2 Natural vegetative reproduction.	Understands how plants reproduce asexually.	• Brings mature suckers, corms, runners, stolons, stem cuttings, bulbs, and rhizomes then show them to learners. • Tasks them keenly observe the plants and discuss what part each use to give rise to new similar plant. • Asks individual Learners to identify and explain how the above plants reproduce. • Encourages them to share the ideas to fellows in different groups.	• Sugar canes • Onions/garlic • Cocoyam • Banana/pineapple sucker • Irish potatoes • commelina • St. Bernard biology book 4 page 2-4	
2.1 Artificial vegetative reproduction		• Distributes text books to different groups bearing images of artificial forms of vegetative propagation. • Guides learners to use text books study the pictures to understand how humans have used knowledge to form new plants from existing ones • Tasks individual learners to make clear notes about the ways of making new plants using human knowledge. Also giving examples in each case	• St. Bernard biology learners' book 4 page 5. • Grafted mango/orange • Printed images/videos of layering, stem cuttings, marcotting etc.	
2.2 Commercial application of vegetative	Understands that asexual reproduction in plants has important commercial	• Displays a video/pictures showing crops commonly and extensively grown in different parts of Uganda • Probes learners to identify crops, and give regions	• St. Bernard biology book 4 page 6-7 • Computer	

reproduction.	applications	where each plant is extensively and commonly grown in Uganda. - In pairs, Learners discuss contribution of a sexual reproduction to the economy of Uganda. - Tasks individual learners to write a public notice encouraging members of societies in Uganda to continue/practice carrying out vegetative propagation.	- Television - Charts - projector	
3.1 Advantages and disadvantages of vegetative reproduction.		- Groups learners into two sides (i.e. boys and girls day and boarding students etc.) - Guides them to debate about advantages and disadvantages of vegetative reproduction. - Guides Learners in their groups to discuss and choose at most 4 representatives. - Tasks learners to write the views pointed out by representatives in their books and also raise points of views as per debate rules and regulations. - Encourages learners to carryout group project during their free time of term. (i.e. in groups of 5 – 10, learners carry out a 3 – week group project using any asexual method to produce a new offspring) Activity of integration	- St. Bernard biology book 4 page 8. - Plain papers - Chalk board - Chalk - Pens. - Stapling machine	

Theme: Reproduction in organisms

Topic: Sexual reproduction in plants

Competency: *The learner understands that a flower is a specialized organ in which all events of a plant's sexual reproduction occur leading to formation of an embryo located in a seed.*

3.2 Flower structure.	Recognize flower structures that are involved in the processes of pollination, fertilization, fruit and seed development and outline their functions.	• Brings different flowering plants (especially dicotyledonous and monocotyledonous plants) to class and asks learners to observe them i.e. a maize and a bean plant) • Tasks learners think of other flowers at their homes and use these to reflect on observations made and asks questions for them give their ideas. • Provides different flowers of plants, learners text books or displays a video showing different flowers • Tasks Learners in their groups to critically examine and discuss the flowers to identify parts which are involved in pollination, fertilization, seed and fruit formation also discuss the functions of parts identified and present their findings to the rest of class. • Guides individual learners draw and label a flower, pollen grain and an ovule	• St. Bernard biology book 4 page11&12 • Monocotyledonous and dicotyledonous pant flowers (hibiscus, elephant grass, morning, maize, glory flowers etc.)	
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4.1 Pollination, fertilization and fruit formation.	Know the processes of pollination, fertilization and fruit formation.	• Provides Learners with charts bearing diagrams /listen a video showing cross and self- pollination; another one showing processes of fertilization and fruit formation • Tasks learners to brainstorm and give reasons why insects in home areas visit some flowers. • Probes learners to define the terms pollination, fertilization and fruit formation. • Tasks Learners to discuss on how pollen grains reach the stigma of flowers. (Elaborates characteristics of wind and insect pollinated flowers) and types of pollination (drawings of and factors favoring each type inclusive.) • Guides Learners to write on the events that occur during double fertilization in a plant. • Through brain storming, teacher tasks learners to write a statement a on what happens to parts of a flower after double fertilization.	• St. Bernard biology book 4 page 13-16. • Videos / television • Charts • Fountain biology book 4 page 10-12	
4.2	Understand the differences between cross and self-fertilization and the advantages of	• Introduces the lesson by making an arecarp of previous lesson and displays a video showing cross and self-fertilization in plants/demonstrates the processes.	• Fountain Biology book 4 page 9 • Video	

Differences between self and cross-fertilisation	each method.	- Asks learners to reflect on the video and answer the question asked. (I.e. how are two processes in the video different?) - Tasks individual learners to produce a table comparing the two processes; after they share their findings with the rest of the class.	- Projector - Charts/Manilas - Television - Projector	
5.1 Seeds and fruits	Differentiate between seeds and fruits structurally and functionally.	- Guides learners to use 5 minutes move around the school compound to collect fruits and seeds seen. - Brings and shows learners different seeds and corresponding fruits of the same plant. (I.e. bean plant and use these to state the common features). - Distributes them in class and asks learners to discuss their differences in in terms of structures and functions. - Probes learners to use fruits and seeds given to define terms “fruits and seed” - Individual learners draw and label seeds and fruits from the same plant then write their comparison in a table.	- St. Bernard biology book 4 page 17. - Bean fruits/seeds - Cow peas seeds/fruits etc	
5.2 Importance of dispersal	Understand the importance of dispersal.	- Narrates scenarios of how different seeds and fruits scatter from the parent plant to new areas. - Tasks learners to reflect on the scenario and explain verbally the usefulness of the context in the scenario.	- St. Bernard biology book 4 page 26. - Manilas/ flip charts - markers	

		Tasks individual learners to write harmonized work in their books.		
6.1 Types of fruits	Recognize the structures and types of fruits and seeds and relate their structures to their methods of dispersal.	- Through research using learner's books/internet/home environment, learners mention different examples of fruits. - Distribute different flowers and fruits to groups of 5-10 learners. - Tasks learners to classify the fruits into 4 categories and then relate these categories to 4 types of fruits i.e simple, multiple, aggregate, and accessory types of fruits. - Guides students in studying the simple fruits provided, and classifies them according to their characteristics. (should also draw and label different simple fruits in their books (here longitudinal and transverse sections of given fruits can be made).	- St. Bernard biology book 4 pages 18-23. - Mature fruits of; - Castor oil/okra - Bean pods - Desmodium - Nut - Bidens pilosa - Tridax - Orange/mango - Etc.	
6.2 Modes of fruit dispersal.		- Displays a video showing different modes of seed dispersal. tasks them to reflect on the video and discuss different modes of fruit dispersal as they are asked questions. - Tasks individual learners to write an essay about modes of dispersal in their books.	- Mature fruits of; Castor oil/okra, Bean pods, desmodium, Nut, Bidens pilosa, Tridax, Orange/mango - St. Bernard biology book 4 pages	

			25&26.	
7.1 Activity of integration	Activity of integration	Individually, the learner attempts the activity of integration assessing him /her to see the level of achievement on the learning outcomes.	- Question papers - Stapling machine - Wall clock - Stapling wires - answersheets	
Theme: Reproduction in organisms Topic: Sexual reproduction in humans. Competency: The learner understands that sexual reproduction involves two parents with a specialized reproductive system.				
7.2 Male and female reproductive systems.	Understand the structure and functions of male and female reproductive system in humans	- Displays images of pregnant ladies, and those with babies and then asks learners to give an overview of how children are born. Learners can express how young babies are born by talking about the crucial body parts which are involved in this. - In groups learners examine charts of male and female reproductive systems. Learners discuss the functions of parts labelled (for both male and female reproductive systems) - Individually learners draw and label the two systems and present the functions in table form as appropriate.	- Charts, - Fountain biology textbook page 18&19 - markers, - Elite Biology Learners book 4, page 19 - fountain biology bk.4 page 18-19. - Printed images.	

8.1 Human Menstrual Cycle.	Understand the changes that take place during the menstrual cycle.	• Groups learners and provides drawn/ printed charts/textbooks showing hormones which are responsible for the menstrual cycle • Tasks learners in their groups/pairs to mention body changes which appear beginning at the age of 13 in girls and then 15 in boys. • Instruct learners to mention the hormones responsible for the menstrual cycle. • In pairs, learners examine graphs showing monthly variation in LH, FSH, Oestrogen and progesterone. • Asks Individual learners to explain through an essay the events of hormones in the graphs. • Share their findings with the rest of the class.	• Fountain biology textbook 4 page 21 • St. Bernard biology textbook 4, page 31 and 32 • Charts • Markers • Cell/masking tapes	
8.2 Fertilization and development of zygote.	Relate the male and female gametes.	• Provide learners with charts/textbooks bearing both sperm and ovum cells. • Learners observe charts, identify parts of each gamete, and draw and label them. • Individual learners write the functions of each part.	• Charts • Markers • St. Bernard biology book4, page 33	
	Appreciate the process of fertilization of an ovum and the development of the zygote up to birth.(u)	• Displays a video using a projector or television/textbooks about the fertilization process and stages of development of the zygote up to birth. • Learners reflect on the video and discuss the events that occur during fertilization. • Individual learners outline the events of the	• Video clip • Projector • Internet • Chalk • Dust	

		process.	• Chalkboard. • Fountain biology book 4 page 35.	
9.1 Antenatal medical care.	Know the role of the placenta during pregnancy.	• Gives learners charts showing the placenta connected to the fetus • Tasks learners to study it carefully, draw and label it. • Gives individual learners the activity of stating the role of placenta assesses and gives feedback.	• Charts • St. Bernard biology book 4 , trs guide page 57.	
	Understand the importance of antenatal medical care (u, s).	• Gives a simple speech/ displays images about pregnant mothers. • Tasks learners to brainstorm about antenatal care as the teacher probes questions to them. • Instructs the learners individually to discuss the importance of antenatal medical care.	• Antenatal medical chart • Printed images • Chalk • Chalkboard.	
9.2 Care for baby after birth, teenage pregnancy and abortion.	Understand aspects of care for the baby after birth(breastfeeding, balanced diet, immunization and hygiene) Recognize the health risks	• Instruct learners to look at images in the textbooks provided and interpret them to know the care given to babies. • Tells learners to do a role-play/ explain verbally about the care mothers give to their children after birth. • Tasks individual learners to write the about care reflected above. • Asks learners to listen to visiting health personnel' s talk and thereafter watch video clips about	• St. Bernard Textbook 4, page 37. • flip charts • markers • Video clips • Projector • Internet	

	/complications associated with early/teenage pregnancy and abortion.	health complications associated with teenage pregnancy and abortion. • Encourage learners to be attentive and ask questions after. • After the talk, ask learners in their groups to make a poster about health complications associated with teenage pregnancy and abortion.	• Manilas	
10.1 Birth control methods	Identify the common birth control methods in Uganda and give the biological principle they employ and their effectiveness(u)	• Using photos downloaded, learners are tasked to carefully observe and define what birth control is. • From the photos, learners identify the different birth control methods, and discuss their effectiveness and the most effective methods for different age groups. • Tasks individual learners to give advantages of each birth control method.	• Downloaded photos • Fountain Biology learners book 4, page 40 • St Bernard learners book 4 ,page 39 • Elite Biology learners book 4, page 27	
10.2 Reproductive abnormalities	Identify and explain common issues associated with reproductive systems(u)	• Invite medical personal personnel to discuss reproductive abnormalities in men and women. • In pairs learners reflect on the medical personnel's discussion, discuss the reproductive abnormalities • Individual learners summarise the abnormalities in a table form.	• Charts, • markers	
11.1 STIs	Knows the causes, signs and symptoms and understands the mode of transmission of named STIs	• Plays a video clip on different STIs and asks learners in class to attentively watch them. • Guides them to reflect on the video and discuss the causes, signs, symptoms, and mode of	• Projector • Computer • Television	

	(Syphilis, gonorrhoea, candida, HPV, Hep B, and HIV/AIDS) (K,U, V)	transmission of STIs.	• animated videos	
11.2 Prevention measures for STIs	Appreciate the preventive measures for the named STI's	• Carries out a guidance and counselling speech in not more than 7 minutes on the importance of abstinence from sex • Tasks learners in their groups to identify other preventive measures to curb mentioned STIs. • Guides and tasks learners to write a poem about ways of preventing any two STIs.	• Charts • Markers • pens • Notebookss.	
12.1 CHALLENGES OF HIV/AIDS	Identify the challenges faced by people living with HIV/AIDS and how to overcome them.	• Tasks learners to listen or recite a poem/song "alone and Frightened by Philly Bongole Lutaaya" • Guides learners in groups to discuss the discrimination and stigma portrayed in the song. • Task individual learners to outline the significance of the song.	• Robot • flash • written poem	
12.2	Activity of Integration	Tasks learners to use the provided materials and individually attempt the activity of integration	Stationery.	

SUBJECT: BIOLOGY **CLASS** S.4 Term: 2 NAME OF

TEACHER:

Time allocation: 4 periods per week (2 doubles)

Theme: genetics Topic: inheritance Competency: The learner appreciates the transmission of characteristics is from parents to offspring's through a process known as heredity.				
1.1 Cell division i) mitosis	Understand the process of cell division.	- Tells learners to think of any person who has ever gotten a cut in their community but he/she heals. - Instructs learners to view an animated video of mitotic cell division. - Thereafter asks them discuss in their groups, the definition of the term mitosis and its process as reflected in the video above. - Individual Learners are tasked to write the process in their books while making clear diagrams.	- Video clip - Projector - Computer - internet	
1.2 ii) meiosis		- Provides a video/ text books with the process of meiosis. - Guides them on how to explain the process occur in their groups. - Tasks individual learners to write the process and also suggest the importance of the process in their books.	- Video clip - Projector - Computer - internet	
2.1 Terms used in inheritance.	Understand the concept of inheritance using genetic diagrams.	Asks learners to read the text in learner's books / think of resemblance and difference in appearance among people in the same family members and connect this to heredity and genetics.	St. Bernard bk. 4 page 50	.

		- Using the internet or text books provided, learners research and about the terms used in genetics as they attempt activity 4.3 in St. Bernard. - Learners share their findings to their fellows in class.		
2.2 Monohybrid inheritance		- Displays pictures of flowers and animals of one color, pink flowers and roan cows. Then he asks to brain storm and tell how those organisms with different colors. Here the teacher emphasizes learners to demonstrate their ideas. - Teacher uses learners' views to demonstrate monohybrid inheritance in organisms. - Gives more questions and tasks them to work it out. - Explains a bout co-dominance and incomplete dominance.	- Pictures showing black, red and roan mammal. - Red, white and pink flowers. - Chalk - Chalk boar - Dust.	
3.1 Sex determination	Understand and explain sex determination in humans.	- Avail learners in their groups with two coins. One coin having a side labelled x and another side labelled y. Another coin with both sides labelled x. - Asks learners in groups of 5-10 to attempt activity 4.6 in St. Bernard book 4 page 54 – 55. - Task individual learners to present their findings to the rest of class.	- Labelled coins - Plain papers - Flip charts - Markers - Chalk board. - St. Bernard biology book 4page 54 – 55	

3.2 Sex Linkage	Understand and explain sex linkage in humans.	- Displays printed images of people with sex linked genes, asks them to observe. - Asks learners to also read text in text book about the sex linkage in humans and interpret the picture on page 58. - Explains more about sex linkage concept. - Tasks individual learners to show how a hemophiliac person come about.	- St. Bernard biology page 58. - Printed images	
4.1 Sex linkage AOI		- Makes a recap of previous lesson and tasks learners to answer the questions as provided. - Tasks learners to give features of albino people and gives a question where they are supposed to show how these are born. Tasks learners to attempt the activity of integration.	- Chalk - Dust - Chalk board - stationary	
Theme: genetics Topic: variation and selection Competency: the learner understands that variation is as a result of changes in genetic make up of an organism				
4.2 types and causes of variation	Appreciate that variation in organisms is due to external and internal factors, and that mutations can be beneficial, harmful or neutral.	- Guides learners to look at their fellow students to know difference in size, weight skin color tongue rolling (if there are twins / students of the same parents in a school – these can be studied about). - Tasks learners to write a report about their findings as he asses them. - In groups, learners use internet and text books to explain factors that lead to variations in organisms.	Note books. St. Bernard biology text book 4 page 62 – 64.	

5.1 Mutations and genetic disorders	Identify diseases associated with genetic disorders e.g sickle cell anaemia, albinism, and Down's Syndrome	• Provide learners with computers well connected to internet / research about the causes and effects of albinism and sickle cell; • Allow learners to share their findings to the rest of colleagues in class.	• Computers • Internet • Projectors	
5.2 Natural and artificial selection.	Understand the concept of natural selection as a mechanism of evolution. Understand the use of artificial selection in selecting breeding.	• Introduces the lesson by asking learners to watch a video about feeding relation of animals in a game reserve • Tasks learners in their groups to give reasons of why some species of organisms become extinct and also why organisms of the same species differ in different time lines. • Tasks individual learners to share their findings. • Learners visit an agricultural facility, or listen to a visiting agricultural extensive officer, and carry out research in pairs and learn about the use of genetics in plant and animal breeding.	• Projector • Video clip • Computer • Manilas • Markers	
6.1	Activity of Integration	Tasks learners to use the provided materials and individually attempt the activity of integration	Stationery.	
Theme: interrelationships Topic: concept of ecology. Competency: The learner understands the concepts of communities, habitats, and ecosystems.				

6.2 Concept of communities, habitats and ecosystem	Know the meaning of the term ecology. Understand the concepts of communities, habitats and ecosystems.	- Provides and guides learners to watch a video/ biology text books/ internet about life in a game park/game reserve; water; deserts, mountains and plains or life in a forest. - Listen carefully and reflect on the videos to make a report. (Report must include explanation of the terms commonly used in study of concept of ecology, and factors involved in the balance of ecosystems). - Individual learners orally answers the questions.	- Video clip - Internet - Projector - Computer - Magazines - newspapers	
7.1	Activity of Integration	Tasks learners to use the provided materials and individually attempt the activity of integration	Stationery.	
THEME: INTERRELATIONSHIPS TOPIC: FOOD CHAINS AND FOOD WEB. Competency: the learner appreciates the interdependence of organisms in a given ecosystem.				
7.2 FEEDING RELATIONSHIPS IN ECOSYSTEMS	Understand the feeding relationship in an ecosystem and express those using food chains, webs, and pyramids.	- Introduces the chapter by clearly stimulating learners with pictures of different organisms and what they feed on. - In groups study a local ecosystem through field visits and observe, identify and record the organisms. (the learners note what food sources the animals rely upon; define food chains and webs; - Tasks learners to make an individual report.	- Printed images - Cameras - Manilas - Markers - St. Bernard text book 4 page 83 and 84. - Fountain biology	

			book 4 page 50-51	
8.1 TROPHIC LEVELS		Tasks them to study the report from previous lesson to draw simple food chains then develop these into food webs)	- Flip charts - markers	
8.2 Ecosystem pyramids.		- Tasks learners in their groups to use text books to identify the trophic levels of organisms and thereafter develop the food chain and food web pyramids also explain their nature. - Individual learner makes notes in his/books which is assessed by the teacher.	St. Bernard text book 4 page 85-87. Fountain biology book 4 page 52	
9.1 Carbon cycle	Appreciate the organisms and processes involved in carbon cycle , and its role in maintaining the carbon dioxide balance in the atmosphere	- Distributes the charts showing the carbon cycle/provides a video for watching about carbon cycle/ text books having carbon cycle. - Tasks learners in pairs to study it carefully. Then discuss given charts to identify and describe the processes involved. - Individually learners draw and show different processes involved in carbon cycle	St. Bernard book 4 page 87. Fountain biology book 4 page 53 - Charts - Markers - Video clip - Projector - computer	

9.2		Guides learners to use textbooks and identify different organisms involved in processes of carbon cycle. Then state the role of each organism in maintaining the carbon dioxide balance in atmosphere	- Charts - markers	
10.1	Activity of integration	Activity of integration	Stationery	

SUBJECT: BIOLOGY **CLASS** S.4 Term: 3 NAME OF
TEACHER:

Time allocation: 4 periods per week (2 doubles)

Theme : interrelationships Topic: associations in biological communities Competency: The learner appreciates that organisms naturally interact in different ways with one another in habitat.				
1.1 Types of competition in ecosystem.	Know what competition is and describe how organisms compete in nature.	- Play a video in class showing competition in an ecosystem and - Pause questions to individuals in groups. - Chosen learners provide ideas about types and importance of competition.	- Questionnaire - St. Bernard biology book 4 page 90 - Fountain biology book 4 page 56	

			• Video clip, Projector.	
1.2 Prey – predator relationship.	Differentiate prey from predators and describe the prey – predator relationship.	• Introduces the lesson by narrating a story about the relationship between rabbits and fox in the bush and asks learners to listen well.(as the number of rabbits decreases , the number of foxes also decrease) • In groups of 3–5, tasks learners to define prey and predator. • Brain storm on more examples of prey – predator relationship • Draw graphs/curves showing prey – predator relationship; and describe this relationship.	• St. Bernard biology book 4 page 91. • Fountain biology book 4 page 57 and 58. • Graph books/papers • Markers • Flip charts.	
2.1 Symbiosis	Understand symbiosis, mutualism, commensalism, and parasitism and appreciate their roles in an ecosystem.	• Introduces the subtopic by giving learners textbooks; they look at different images showing relationships among different organisms. • Tasks learners to observe images carefully and come up with definitions of symbiosis, mutualism, commensalism and parasitism. • Asks learners through question and answer the other examples of relationships above. • Individual learners write the roles of each relationship above in their books, calls the teacher for marking.	• Fountain biology book 4 page 58 and 59 • St. Bernard biology book 4 page 92 and 93. • Printed images	

2.2 Parasites and vectors.	Recognize the role of parasites and vectors in the transmission of common diseases (nagana, malaria, bilharzia and sleeping sickness)	• Brings tick and tape worms in class, asks learners to observe them then define the terms parasite and vector. • Tasks Learners in their groups to visit a nearby health/agricultural facility or listen to a visiting health/agricultural officer, carryout research to find out about common parasitic diseases, how they are transmitted and how they can be prevented. • Tasks learners to make a report about the study and present their work to teacher for assessment.	• Ticks • Tape worms • Pens • Note books • Charts • Markers.	
3.1 Adaptations of parasites.	Know the adaptations of parasites to their mode of life.	• Brings tick and tape worms in class, asks learners to observe • Guides learners to develop and present a role play/ a poem showing how different parasites are adapted to their mode of life and how humans try to prevent their transmission. • Individual learners makes a write up about the role play/ poem done.	• Ticks • Tape worms • Pens • Note books • Charts • Markers. • Flash/memory card.	
Theme: INTERRELATIONS				

Topic: Humans and the Natural Environment competency: the learner appreciates that uganda has different natural features, that our activities have an impact on these resources, and recognises the reasons why countries have committed to global sustainable development.				
3.2 Sustainable development goals.	Understand there is a worldwide focus on sustainability and importance(u).	• Displays a video on sustainable development goals. • Instructs learners to reflect on the video following the teachers ' guidelines. • Tasks groups to refer to activity 9.1 in the learner's book. • Individual learners are tasked to write a report on SDG'S	• Plastic bottles • Polythene bags • St. Bernard biology book 4 page 95-97 • Internet • Fountain biology book 4 page 63. Internet, pictures	
4.1 Natural resources in Uganda.	Know and give examples of natural resources found in Uganda .(k,u)	• Provides learners with pictures/newspapers showing natural resources. • Tasks learners through pair-share to discuss and identify the natural resources. • Individual learners write an essay on the natural resources pointing out the types, and their importance in Uganda as a country.	• Pictures, • Newspapers • Video clips • Television • Internet. • Fountain biology book 4 page 64&65 • St. Bernard book 4 page 97	

4.2 Factors which impact ecosystems.	Appreciate and describe natural factors and human influences that may impact ecosystems and suggest how to preserve the natural environment for all living things. (u,s,v)	• Displays videos/images showing examples of human activities (human influence) on the natural resources, enrichment • Learners in their groups are tasked to reflect and discuss the human activities displayed and point out their dangers and mitigations • Tasks individual learners to write an article to be published in new vision about how natural and humans factor impact on ecosystems.	• Video clips • Printed images • St. Bernard page 98&99	
5.1 pollution	Understand the sources, effects and control of air, land and water pollution. (u,s,v)	• Group learners into manageable groups, take learners to a nearby grinding sawmill or display video showing what happens in an environment of a factory or industry. • Refer learners to pictures on the learner's book st Bernard book 4 page 99-100 to bridge their understanding on pollution. • Asks learners to suggest the kinds of pollution noticed in the sawmill or the displayed video. • Tasks individual learners to write an article entitled effects of pollution.	• Pictures, • Newspapers • Video clips • Television • Internet. • Fountain biology book 4 page 65&66 • St. Bernard book 4 page 97	
5.2 Renewable and non – renewable resources		Take learners to nearby garbage collection sites or garbage deposits or display broken, used plastic materials. • Refer learner to activity 9.7 on the learners book st Bernard book 4 page 103.	Polythene bags • Plastic bottles • Polythene bags • St. Bernard biology book 4 page	

		Tasks individual learners to write an essay indicating clearly where effective re-use and re-cycling takes place in Uganda, and what should be done to improve the process.	99-101. Internet Fountain biology book 4 page 66-67.	
6.1 Activity of integration	Activity of integration	Activity of integration	Answer sheets Stapling machine Stabling wires.	
6-12	Weeks 6-12 should be used to prepare for and administer UNEB Examinations			